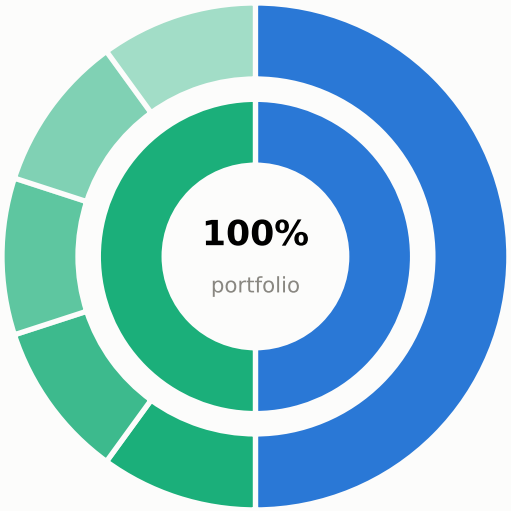
STRATEGY HANDBOOK

Core & Shield

How it works, why it exists, and how to judge it

Version 2.0 · 14 October 2026 · written before the first trade (1 October 2026); v2.0 adds the QQQM trend Core, live from 2 No

> **In one sentence:** half the money is a growth **Core** (the Nasdaq-100, held only while it is above its 10-month average), the other half is run by a rules-based model (the **Shield**) that each month picks the strongest, calmest and least-crowded of 12 ETFs, or retreats to cash. The goal is to beat a 60/40 on return with much smaller losses when markets fall.



- **Core · 50%** · growth, with a trend filter
  - QQQM · Nasdaq-100 while it trends up
  - T-bills (BIL) when it is below its 10-month average
- **Shield · 50%** · RAAM, re-picked monthly
  - 5 slots × 10% each, chosen from 12 ETFs
  - a slot goes to cash (BIL) if its ETF is weaker than cash
  - Bitcoin (IBIT) never more than 5% of the whole portfolio

Backtest, Feb 2015 – Sep 2026, after trading costs (Bitcoin before 2024 simulated; see section 6)

Return per year	Sharpe vs T-bills	Calmar ratio	2022 (bear market)
11.9%	**1.01**	**1.02**	**−4.9%**
60/40: 9.2%	60/40: 0.73	60/40: 0.48	60/40: −15.2%

At a glance	
Account	Alpaca paper trading account, starting value $100,000 (move to IBKR paper planned)
Trades	Once a month: signal after the last trading day's close, orders the next morning
Run by	Claude Code cloud routines; the maths and orders are a fixed Python script
Alerts	Discord: daily health line, monthly PDF report, any halt
Status	Live since 1 Oct 2026; current Core from 2 Nov 2026, when the 12-month evaluation restarted

1 The idea

Why this strategy exists, where it comes from, and what it is (and isn't) for.

The problem it solves

A 60/40 portfolio is the standard balanced benchmark, but its losses in a bad year are large: **−15% in 2022** and about **−19%** at its worst month-end in our test window. Growth assets compound faster but fall further. Tactical rotation models sidestep slow bear markets but lag in bull markets.

Core & Shield gives each half one job. The **Core** is pure growth, the Nasdaq-100 through QQQM, with one simple exit: it moves to T-bills when the index is below its 10-month average. The **Shield** is a rotation model that behaves like a shock absorber. Because the Shield already defends, the Core doesn't need bonds. Together they aim to **beat a 60/40 on return with much smaller drawdowns**.

Where it comes from

The Shield is the **Ranked Asset Allocation Model (RAAM)** by Gioele Giordano, which won the 2018 Charles H. Dow Award from the CMT Association. RAAM ranks a diversified set of ETFs on momentum, volatility and correlation each month and holds the best five, moving to cash when nothing is trending up. The Core's trend filter is the 10-month moving-average rule studied by Faber (2007).

Our version is **not** the paper's version. The paper's headline (+638%, Sharpe 1.94 over 2004–17) was in-sample, before costs, and could not be reproduced. We resolved the paper's seven ambiguities, made three safety upgrades, removed a factor that hurt results, added a growth Core with a trend filter, and added a small, capped Bitcoin sleeve. Section 9 lists every change and the evidence.

What it is designed to do

- **Grow, then step aside.** The Core rides the Nasdaq-100 in uptrends and sits in T-bills in long downtrends, as in 2022.
- **Defend with the Shield.** The Shield holds what is trending, calm and diversifying, and moves slots to cash when trends break.
- **Follow rules, not opinions.** The allocation is computed by a script. Nobody, including Claude, overrides it because of news or a market view.
- **Trade rarely.** One rebalance a month, about 21% of the portfolio turned over each time.

What it is not

- **Not diversified away from US tech.** When the Core is on, half the portfolio is the Nasdaq-100. A tech-led bust is the main risk (section 8).
- **Not a crash-proof product.** A sudden one-day crash hits before any monthly signal can react; the circuit breaker limits the damage but cannot prevent it.
- **Not a Bitcoin strategy.** Bitcoin is one candidate of twelve and is capped at 5%.

Who a strategy like this suits

Investors who want more than a balanced portfolio's return but are judged on drawdowns and risk-adjusted return, and who accept a concentrated growth engine in exchange.

> **Its personality in one line:** a growth engine with two brakes. It expects to beat 60/40 in most years and to lose much less in long bear markets, but it is exposed to sudden crashes and to a tech-led bust.

2 How the portfolio is built

Two halves, 13 possible holdings, one cap.

Half	Weight	Holdings	How it changes
Core	50%	QQQM (Nasdaq-100), or BIL when the trend filter is off	Every month: held while QQQM's month-end close is above its 10-month average, else T-bills.
Shield	50%	5 slots of 10% each, from 12 candidate ETFs, with BIL (cash) for failed slots	Re-chosen every month by the RAAM rules in section 3.

The two halves simply add up. When the Core's trend filter is off and the Shield's picks fail the cash test, the whole portfolio can sit in **BIL**, a 1–3 month Treasury bill fund that earns the short-term interest rate. In the backtest the Core was off in about 1 month in 5.

The Shield's 12 candidates

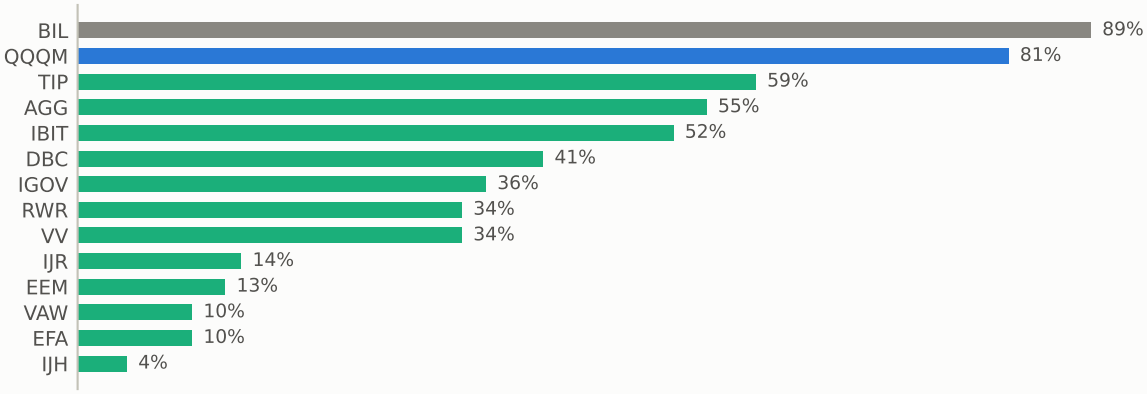
Eleven ETFs from Craig Israelsen's **7Twelve** diversified portfolio, plus spot Bitcoin. Together they span stocks, real estate, commodities, bonds and crypto, so there is usually something trending up.

Asset class	ETF	What it holds
US stocks	VV	Vanguard Large-Cap: large US companies
	IJH	iShares Core S&P Mid-Cap: mid-sized US companies
	IJR	iShares Core S&P Small-Cap: small US companies
International stocks	EFA	iShares MSCI EAFE: developed markets outside North America
	EEM	iShares MSCI Emerging Markets
Real estate	RWR	SPDR Dow Jones REIT: US real estate investment trusts
Commodities / resources	DBC	Invesco DB Commodity Index: energy, metals, agriculture
	VAW	Vanguard Materials: mining, chemicals and materials companies
US bonds	AGG	iShares Core US Aggregate Bond: the broad US bond market
	TIP	iShares TIPS: inflation-protected US Treasuries
International bonds	IGOV	iShares International Treasury: developed-market government bonds
Crypto	IBIT	iShares Bitcoin Trust: spot Bitcoin. **Capped at 5% of the portfolio**
Cash (not ranked)	BIL	SPDR 1–3 Month T-Bill: where failed slots and any IBIT excess go
Core only	QQQM	Invesco Nasdaq-100: the Core's growth holding (not a Shield candidate)

The Bitcoin cap

A normal Shield slot is 10% of the portfolio. If IBIT is picked, it only gets **5%**; the other 5% goes to BIL. The cap is enforced by the script after the ranking, so the portfolio never starts a month with more than 5% in Bitcoin (between rebalances it drifts with Bitcoin's price). A Bitcoin crash of −80% while it is held costs roughly 3–4% of the portfolio.

How often each ETF was held (backtest, % of months)



Blue = the Core (QQQM, held whenever its trend filter is on). Grey = cash. Green = Shield picks.

3 The monthly decision

Eight mechanical steps, computed by the script after the last trading day's close.

1 Measure — M, V, C for all 12 ETFs at month-end

2 Rank — 1 (best) to 12 on each factor

3 Score — average of the three ranks

4 Pick 5 — the five lowest scores, 10% each

5 Cash test — a pick weaker than cash goes to BIL

6 Add the Core — QQQM 50% if trending, else BIL 50%

7 Cap Bitcoin — IBIT ≤ 5%, excess to BIL

8 Trade — next morning, then hold all month

The three factors

Factor	Measured as	Best rank goes to	Why it matters
M · Momentum	Price change over the last 84 trading days (about 4 months)	Highest momentum	Assets that have been rising tend to keep rising for a while
V · Volatility	Daily price swings (EWMA, λ = 0.94, annualised, smoothed over 10 days)	Lowest volatility	Calmer assets give smoother returns and smaller losses
C · Correlation	Average correlation of daily returns with the other 11 candidates over 84 days	Lowest correlation	Assets that move differently add real diversification

Score and selection

Score = ⅓ × Rank(M) + ⅓ × Rank(V) + ⅓ × Rank(C) − Momentum ÷ 1000 (lower is better)

- Each candidate is ranked 1 (best) to 12 on each factor; ties share the average rank.

- The tiny momentum term only breaks exact ties, in favour of the stronger trend.
- The **five lowest scores** are picked. Each gets 20% of the Shield, which is 10% of the portfolio.
- **Cash test:** a pick keeps its slot only if its 84-day momentum beats cash's own (BIL's) 84-day momentum. Otherwise that slot goes to BIL. If nothing beats cash, the whole Shield (50%) sits in BIL.
- A fourth factor from the paper (an ATR trend/breakout signal) is still calculated and logged, but **no longer scored**: three separate backtests showed results were better without it.

Worked example (illustrative numbers)

Suppose the five lowest scores are TIP, IBIT, DBC, AGG and VV, cash momentum is +1.3%, VV's momentum is −1.0%, and QQQM is above its 10-month average:

Step	QQQM	AGG	TIP	DBC	IBIT	VV	BIL	Total
Shield picks (10% each)		10%	10%	10%	10%	10%		50%
Cash test: VV −1.0% < cash +1.3%		10%	10%	10%	10%	0%	10%	50%
Add the Core (trend filter on)	50%	10%	10%	10%	10%		10%	100%
Cap IBIT at 5%	50%	10%	10%	10%	**5%**		**15%**	100%

The last row is what gets bought on the first trading day and held for the month.

4 Timing and operations

Who does what, and when. Everything runs in the cloud; the GitHub repository is the system's memory.

A month in the life

When (US Eastern)	What happens
Every weekday after the close (17:30 summer / 16:30 winter)	Health check: account value, drawdown, positions vs targets, circuit breaker. One Discord line.
Last trading day, after the close	Research scan first (context only), then the **signal** is computed and saved. Pre-flight check that every ETF is tradable.
Last trading day, 19:00 summer / 18:00 winter	One-page **monthly PDF report** (account vs 60/40) posted to Discord.
First trading day, 10:45 summer / 09:45 winter	**Rebalance:** sells first, then buys, as market orders using fractional shares (whole shares for any ETF that can't be fractional). Fills and slippage logged.
First trading day, after the close	**Monthly review** against the evaluation scorecard.
Rest of the month	Nothing. No stops, no intra-month trades, no reacting to news.

Routine schedules are stored in UTC, so clock times shift by an hour when US daylight saving ends or starts. Both versions are safely after the open or the close. Holidays are handled by Alpaca's market calendar.

Who does what

Part	Role
scripts/raam.py	All the maths, every guardrail and every order. Deterministic: same data in, same answer out.
Claude (cloud routines)	**Operator, not strategist.** Runs the commands, writes the logs, does research for context, sends the Discord messages. It may halt if something looks wrong. It never changes targets, picks, timing or rules.
GitHub repository	The only persistent memory: config, rules, signals, trade log, reports. Every run starts from a fresh copy and pushes its results back.
Alpaca	Paper brokerage account and market data (a move to Interactive Brokers paper is planned after a feasibility check).
Discord	Where every message, report and alert arrives.
The owner	The only one who can change the rules, in an interactive session, with the change and its evidence written into the change log.

Principles

- **A missed rebalance is cheap; a wrong trade is not.** When data is stale, a fund changes, weights don't add up or the broker errors, the system halts and alerts instead of improvising. (Stress tests: filling up to 5 days late made no meaningful difference.)
- **Research never changes the allocation.** The one exception is a structural problem with an ETF (closure, merger, ticker change, trading halt), which is grounds to halt.
- **Keys live only in the cloud environment's variables**, never in files, commits or messages.

5 Safety rails

Checks that sit outside the strategy and can only stop trading, never add risk.

Circuit breaker

- **Trips** when the whole portfolio is **20% below its high-water mark**, or falls **5% in a single day**.
- **Action:** the next morning, everything (Core included) moves to 100% BIL.
- **Re-entry:** automatic at the next monthly rebalance, following a fresh signal, with the high-water mark reset.
- **History:** in the 2015–2026 backtest it tripped once, on the daily rule (12 Mar 2020, a −5.1% day; worst daily drawdown −15.4%), and re-entered at the April rebalance.
- **Trade-off:** in a one-day crash it locks the loss and can miss a fast rebound until the next month. It is a tail-risk net, not a return enhancer.

Checks before every signal and trade

Check	If it fails
Every ETF has a fresh price bar for the signal date	Halt: no signal is saved
Momentum, volatility and correlation are valid numbers	Halt
Target weights add up to 100% (±0.5%); none negative	Halt
Only universe tickers can be traded	Halt; stray positions are left alone and flagged

No single buy above 55% of the account (a full 50% Core buy must fit)	Halt
Account is a paper account (live needs two explicit switches)	Refuses to run
Market is open at order time	Halt; nothing submitted
Each ETF is tradable on Alpaca	Halt; non-fractionable ETFs use whole shares
Buys never exceed available cash	Buys scaled down to fit
Every order ends 'filled'	Halt and alert, with the fills recorded
The same month can't be traded twice	Second run does nothing

Tested failure drills

Automated tests simulate a rejected order, a partial fill, a broker outage, missing, stale and zero prices, an untradable ETF and an unexpected position. All end in a safe halt with no wrong trade. Two small gaps the drills found were fixed on 14 Oct 2026:

- **F1:** if the broker fails halfway through a rebalance, the orders that already filled (and the one that failed) are now written to the trade log, and the run halts for the owner.
- **F2:** a blank or zero closing price anywhere in the lookback window now halts the signal, instead of the ETF being ranked on the previous day's price.

6 Historical results

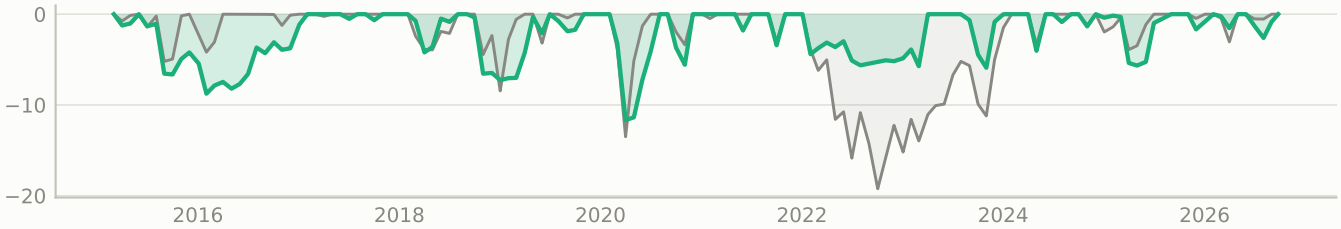
What the rules would have done from February 2015 to September 2026.

How the test works: the live script's own signal code, run at every month-end on Yahoo Finance adjusted prices; fills at the next day's open; 5 basis points of cost per trade; the circuit breaker checked daily. IBIT only exists from January 2024, so before that it is simulated from the Bitcoin price minus IBIT's 0.25% fee. Past results do not guarantee future results.

Growth of $100,000 (log scale)



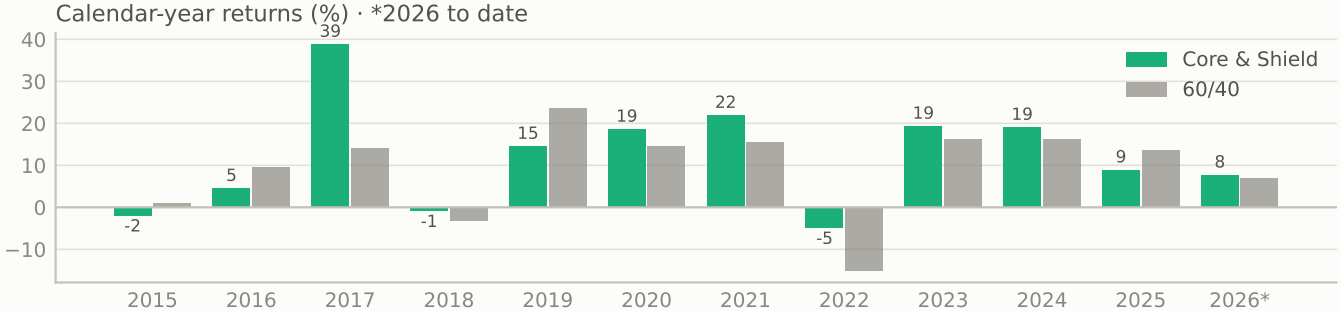
Drawdown from peak, month-end (%)



	Return / yr	Sharpe	Sortino	Calmar	Worst month-end	Worst day	2022
Core & Shield	11.9%	1.01	1.77	1.02	−11.7%	−15.4%	−4.9%
Without Bitcoin	9.0%	0.79	1.28	0.74	−12.2%	−16.1%	−3.8%
Previous (60/40 Core)	9.3%	0.93	1.58	0.84	−11.1%	−14.1%	−8.8%
60/40 (SPY/AGG)	9.2%	0.73	1.11	0.48	−19.2%		−15.2%
SPY	14.1%	0.81	1.24	0.61	−23.3%	−33.7%	−18.0%

Sharpe and Sortino against T-bills (2.0%/yr); at a 0% rate the Core & Shield Sharpe is 1.21 vs 0.93 for 60/40. Calmar = return ÷ worst month-end drop. 2008–2026 without Bitcoin: 8.8%/yr vs 8.5%, Calmar 0.72 vs 0.28; Jun 2008–Jun 2009 about −1% vs 60/40 −17.7%.

Calendar-year returns (%) · *2026 to date



Reading it: ahead of 60/40 in 8 of 12 years, including 2022 (−4.9% vs −15.2%) when the Core's filter moved it to T-bills. Behind in 2015, 2016, 2019 and 2025. The advantage is concentrated in 2020–26; in 2015–19 the previous 60/40-Core design did slightly better.

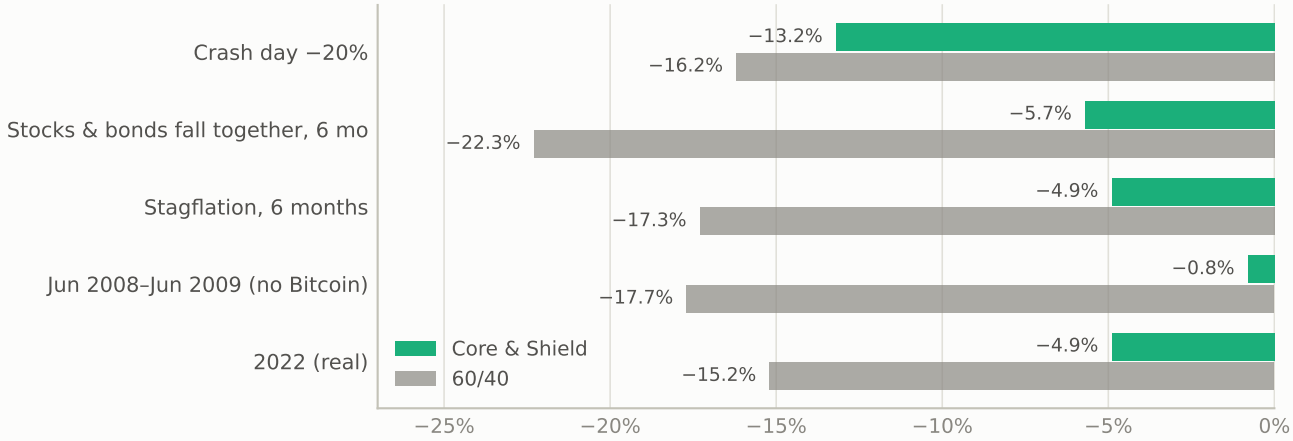
7 Stress tests

How fragile is the result? Rerun 14 Oct 2026 for the QQQM trend Core with backtest/stress.py.

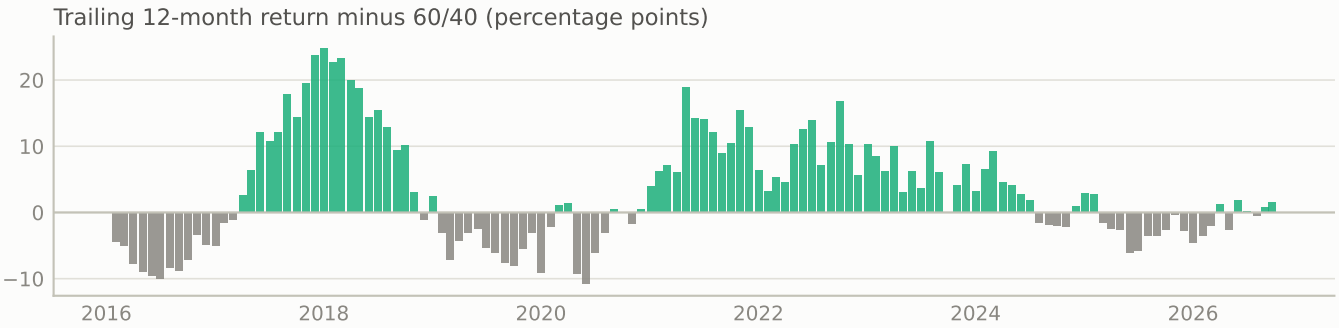
Test	What was tried	Result
Trading costs	5, 10, 25, 50 bps per trade	The edge over 60/40 holds to about 25 bps (Sharpe, rf = 0: 1.09 vs about 0.92) and is nearly gone at 50 bps (0.95).
Settings	13 nearby settings: lookbacks 63–126 days, hold 4 or 6, λ 0.92/0.97, each weight ±10 pp	**13 of 13** still beat 60/40's Sharpe (range 1.12–1.27). Not a lucky peak.
Late fills	Trade 2, 3 or 5 days after the signal	Costs a little (Sharpe 1.20 → 1.08 at 5 days); still ahead of 60/40.
Any single year	127 rolling 12-month windows	Beat 60/40 on return in **59%**, and in **all 14** windows where 60/40 lost money.
Monte Carlo	5,000 simulated 10-year paths (6-month blocks of history)	**89%** chance of a higher Sharpe than 60/40, **80%** of a smaller worst drop. Any single year: 38% chance of trailing 60/40, 13% chance of a loss.

Shocks worse than anything in the test window

Each made-up shock was injected into real prices at six different dates and the whole model re-run, so it reacts as it would live. Averages shown; real 2008 and 2022 added for comparison.



- **Slow crises are where the new Core shines:** stocks and bonds falling together cost about 6% (previous design 10%, 60/40 22%), because the trend filter steps aside.
- **Crash days are the weak spot:** about −13% (previous design −9%). The Nasdaq-100 Core is fully exposed until the monthly filter reacts; the −5% daily breaker then holds cash until next month.
- **Outside the data:** in the 2000–02 dot-com crash the trend-filtered Nasdaq-100 alone fell about 36% before stepping aside, roughly −18% on this portfolio. Plan for a 15–20% drawdown being possible.
- **Bitcoin −80% while held:** about 3 percentage points extra loss, as the 5% cap intends.



Green = ahead of 60/40 over the previous 12 months.

8 What to expect, and how to judge it

The criteria were reset for the new Core before any of its results existed, so results can't move the goalposts.

Strengths and weaknesses

Good at	Weak at
Long bear markets and rising-rate years (2022: −4.9% vs −15.2%)	Sudden one-day crashes: about −13% in a simulated −20% crash day
Beating 60/40 in most years (8 of 12) with smaller drawdowns	A tech-led bust: the Core is the Nasdaq-100 when it's on
Fast recoveries: the worst drawdown took 4 months to recover (60/40: 16)	Choppy markets, where the trend filter can sell low and buy back higher (2015–16)
Robust to settings and late trades; less dependent on Bitcoin	Higher turnover (about 21% a month) and short-term gains in a taxable account

Normal ranges (from the backtest) for judging a live month

Measure	Normal	Unusual / investigate
Monthly return	Between −4% and +4% (81% of months); 66% of months positive	Outside −8.7% to +9.1%
Cash (BIL) weight	About 26% on average; up to 100% when the Core and Shield are both defensive	Cash that doesn't match the rules
Drawdown from peak	Up to about −12% month-end, −15% daily	Worse than −20%
Share of 60/40's loss in its down months	About 72%	Above 85% on average
Core trend filter	Off (Core in T-bills) about 1 month in 5	Off/on not matching the rule
Bitcoin (IBIT)	Held in about half of months, at 5%	Above 5% at a rebalance

Checkpoints

When	Question	Passes when

1 month (early Dec 2026)	Does the new design run?	All operations criteria pass for November: signal and rebalance on time, all orders filled, slippage ≤ 10 bps, holdings within ±2 pp of target, IBIT ≤ 5%, report delivered
3 months (early Feb 2027)	Is it reliable?	Operations pass 3 months out of 3; returns, cash and drawdown inside the normal ranges
6 months (early May 2027)	Does it behave as designed?	Live returns match a backtest of the same months within 0.3 pp a month; defence assessed if 60/40 had 2+ down months
12 months (early Nov 2027)	Is it worth running?	See the verdict rules below

The 12-month verdict

- **Working as designed:** operations pass in 11+ months, behaviour matches the backtest, and the 12-month drawdown is smaller than 60/40's (plus the down-month test, if 60/40 had 2+ down months).
- **Inconclusive:** everything passes but 60/40 had fewer than 2 down months. Normal in a calm year; the defence hasn't been tested yet. Keep running.
- **Not working:** the live account doesn't match the model, or it takes more than 100% of 60/40's losses, or its drawdown is larger than 60/40's in a year when 60/40 fell.
- **Explicitly not a failure:** trailing SPY, or trailing 60/40 in a single strong year.

> **Rule for the evaluation year:** don't change the strategy unless something is broken. A change resets the evidence. The previous 60/40-Core design and other ideas are shadow-tracked instead.

9 Decision history

How the strategy got here, including what was tried and rejected.

Date	Decision	Evidence
23 Sep 2026	Set up pure RAAM from the paper, with defaults for its 7 ambiguities	Paper's +638% claim could not be reproduced
23 Sep 2026	**Tier 1 upgrades:** cash test uses BIL's momentum (was > 0%); BIL replaces SHY as cash; breaker at −20% with automatic re-entry (was −15%, manual)	Safer behaviour at little cost
23 Sep 2026	**Trend factor removed** from the score	Three backtests agreed: better return, smaller drawdown, in both halves of the period
24 Sep 2026	**Core & Shield:** 50% fixed 60/40 + 50% RAAM	RAAM alone lagged badly in bull markets; the blend matched 60/40's Sharpe with far smaller drawdowns
24 Sep 2026	Strategy locked for a 12-month paper evaluation	Owner decision
Sep 2026	Trade the morning after the signal (10:45 ET summer / 09:45 winter)	Month-end close, first day and mid-month compared; no major downside to keeping it

26 Sep 2026	**IBIT added**, capped at 5% of the portfolio	Return 6.5% → 9.3%, Sharpe 0.95 → 1.18, worst drop slightly better; better in every sub-period; about half the gain survives with Bitcoin's long-run return flattened to 0%
26 Sep 2026	Evaluation criteria written and fixed	Before any live result existed
27 Sep 2026	Stress tests run	Robust to costs, settings and timing; worst case about −21%
14 Oct 2026	Fault-drill gaps F1/F2 fixed; shadow tracking of alternatives started	Tests now pass; no strategy change
14 Oct 2026	**Core changed to QQQM + 10-month trend filter** (live from 2 Nov); evaluation restarted	Calmar 0.84 → 1.02 (2015–26) and 0.35 → 0.72 (2008–26, no Bitcoin). Known risks: crash days, a tech-led bust, Nasdaq chosen with hindsight

Tested and rejected, or parked

Idea	Verdict	Why
Swap VV→TLT and VAW→GLD (crisis assets)	Rejected	Better before 2020, worse after; made March 2020 and 2008 worse
Correlation measured against the Core too	Rejected	Worse from 2020 onward
Size positions by inverse volatility	Rejected	Failed from 2020 onward
Heavier momentum weight	Not recommended	Trades defence for return
Bitcoin uncapped (10% slots)	Not adopted	12.7%/yr but more dependent on Bitcoin
Volatility-managed Core	Parked	Shallower losses in every crisis, same Sharpe, about 0.7%/yr less return
Remove VV / swap AGG→IEF (overlap fixes)	Rejected	Worse in both windows; the correlation factor already handles the overlap
Gold interchangeable with Bitcoin	Rejected	Gold crowds Bitcoin out; about −1%/yr
Ranked growth Core; SPY or VTI + trend	Rejected	Lower Calmar or Sharpe than the QQQM trend Core
Gold as a 13th candidate	Shadow-tracked	Planned decision in January 2027; worse 2008 tail
Risk-based sizing, staggered rebalancing	Parked	Candidates for the 6-month checkpoint

Honest assessment

Rated about **70/100** at launch as a defensive balanced strategy with a thin edge. The QQQM trend Core raises the backtested return and Calmar substantially and cuts the dependence on Bitcoin, but it was chosen knowing the Nasdaq-100 led 2010–26, and it adds crash-day and tech-concentration risk. About 11 years of data means a Sharpe estimate is only accurate to roughly ±0.3. The live year from 2 November 2026 is what decides it.

Glossary

Term	Meaning
60/40	60% US stocks (SPY) + 40% US bonds (AGG): the benchmark
Sharpe ratio	Return per unit of volatility; higher is better. 1.0+ is good for a balanced fund
Drawdown	Fall from the highest value reached so far

Momentum	How much a price has changed over a lookback period (here about 4 months)
Basis point (bp)	0.01%. 10 bps = 0.1%
Slippage	Difference between the price when an order was planned and the price it filled at
Circuit breaker	Automatic move to cash after a −20% drawdown or −5% day
High-water mark	The highest account value so far, used to measure drawdown